Zach Murray

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Experience

Senior Software Engineer

One Health Link

Present

October 2024

- Created parallel asynchronous runners handling 30 thousand scenarios each run with Python 3, XDist, and Asyncio to JSON reports to simulate each route of chatbot conversations
- Developed and deployed Mailgun HTML emails and SMS messages from production services to customer emails and phones reaching out to 22 million medical members
- Coded autogenerated Microsoft Copilot API executions from Google Sheets, Microsoft Excel, and CSV files to Sanic and Ruby on Rails servers to handle millions of questions and prompts
- Built auto-generated Playwright scripts that interact and report web portal chatbot deployments on the Microsoft Azure App Service (PaaS) to handle billions of patient interactions
- Created functional annotations with Python decorators to WORA (Write Once Run Anywhere) setting up conversation chatflow on MySQL servers hosted on MS Azure Cloud (IaaS)
- Parsed Draw.io flowcharts from XML with Pandas DataFrames to vertexes and directed edges into ArangoDB (Graph Database System) Docker containers for node and arc analysis
- Calculated the cost of running Linux virtual machines, Kubernetes pods,.and Azure Functions (FaaS) for hosting 10 distributed stations handling on-demand requests
- Updated and maintained SQL and NoSQL database servers to setup safe chatbot states in Dev, Staging, and Production environments for each conversation execution
- Queried MySQL and MongoDB statements for offshore team evaluations of websites, chatbots, and servers states to resolve discrepancies that would be seen by 39 million people
- On call outside hours and weekends with Slack and phone for team members across the globe including China, Brazil, and Ukraine
- Deployed GitHub codebase with Terraform to Dev and Production on Azure Cloud Services

Senior Software Engineer

Tom Sawyer Software

Tom Sawyer Perspectives

August 2024

February 2023

GDIT (General Dynamics IT) | Northrop Grumman (NG) | Federal Contract

- Converted customer-customized proprietary-project to upgrade from Java 8 to 11 by transitioning Maven and in-house packages modernizing aging systems to keep support of global stability
- Developed Spring Security system handling JWTs (JSON Web Tokens) from Keycloak SSO (Single Sign-On) services that are shared within security-clearance air-gapped environments
- Created publishers and subscribers handling data pipeline interactions on Apache Kafka deployments to scale throughput and availability following the Pub/Sub messaging pattern

CGI Inc. | CGI Federal | Federal Contract

- Created SSO authentication with Keycloak, Spring-Boot, and Java to mock DIA DTMI (Defense Intelligence Agency: Data Transformation of Foundation Military Intelligence) access
- Built CDC DB event streaming with Debezium, Kafka, and ZooKeeper to model handling of the DIA MIDB (Modernized Integrated Database) for downstream data ingestion
- Developed alerts and data handling for DIA MARS (Machine-assisted Analytic Rapid-repository System) historical data with AWS Lambda from AWS S3 to DynamoDB and PostgreSQL on RDS

Services

- Created Apache Tinkerpop 3.0 queries in Apache Groovy to import lung cancer data from the Harvard Dataverse into our Neo4j GDB (Graph Database) to export CSVs for file ingestion
- Built GDBs' MVPs of: ArangoDB, ArcadeDB, Dgraph, Janusgraph, Neo4j, and OrientDB with Docker containers for QFD (Quality Function Deployment) HOQ (House Of Quality) evaluations
- Scripted with SQL, GraphQL OpenCypher, Gremlin, and SPARQL query languages for graph traversal of non-tabular nodes and edges for GDBs' performance and feature assessments
- Deployed NoSQL DBs including: Apache Cassandra, Neo4j, and OrientDB for load testing with ICIJ (International Consortium of Investigative Journalists) Offshore Leaks database dump
- Developed toolbar buttons and popups with Spring Boot and Java Swing to allow reordering of tree-view items to continue behavior for rules and their actions of drawing view behavior
- Debugged sorting tree view nodes alphabetically implemented with Java AWT (Abstract Web Tool) and TS custom lists and iterators to improve our link analysis experience
- Coded a React and Node.js POC working with our Spring Framework and Swagger hosting our REST API to develop a renewed frontend and custom-customer experience
- Built set tags actions popups and dialog boxes with TS custom Java descriptor classes and handling files of 4000+ lines to streamline interactions by removing set properties actions

Senior Software Engineer

Ascension

January 2023

November 2021

APDH (Ascension Provider Data Hub)

- Consolidated datasets and scheduled stored procedures in GBQ (Google BigQuery) to ingest from multiple data sources, including AthenaHealth, MD-Staff, and PeopleSoft
- Modeled and deployed deeply and widely nested GBQ dataset tables and views with nests, pivots, and structs from CSVs (Comma-Separated Values) and JSONs
- Created Airflow DAGs in Python generating MDM (Master Data Management) datasets and publishing millions of transactions into Reltio RDM (Reference Data Management)
- Automated daily cron jobs with Apache Airflow's DAGs (Directed Acyclic Graphs) deployed on Google Cloud Composer to maintain Reputation.com information
- Processed individualized asynchronous and distributed computing by utilizing Apache Beam's ParDos and their DoFns to create a fault-tolerant and streamlined REST request response
- Transformed processing functions into generic parallel processing on GCP Dataflow deployed with Terraform to handle thousands of deeply nested key-value structures
- Coded SQL tables, views, and stored procedures in Google Standard SQL on GBQ for MD-Staff, PeopleSoft, and Athena data-sources into Reltio, Reputation.com, and MyChart
- Packaged multilanguage JARs (Java ARchives) of Java, Python, and SQL with Maven and deployed with Jenkins to create reproducible standardized deployments

EMPI (Enterprise Master Patient Index)

- Replicated data extraction from MSSQL DB-Servers to GCS (Google Cloud Storage) Buckets for data ingestion of NextGate EMPI as a data source for CDC (Change Data Capture)
- Engineered data ingestion of pipe-delimited DSVs (Delimiter-Separated Values) with Google Cloud Functions transforming into a RDBMS (Relational Database Management System)
- Automated daily runs of cron-job stored-procedures with Google Cloud Scheduler reconstructing on-premise database-servers over to the cloud dataset-environments
- Remodeled DDL (Data Definition Language) and DML (Data Manipulation Language) into JS (JavaScript) on Google Cloud Functions for dynamic low-maintenance computing
- Engineered and triggered GCP Functions with GCP-Pub/Sub Topic-Messages to handle on-demand API requests across GCP Projects and deployment environments
- Rebuilt Node.js scripts into Python 3 functions to redress CICD (Continuous Integration and Continuous Deployment) complications with Terraform deployments
- Make consolidated views that aggregate data as summations for CEM (Customer Experience Management) incoming outgoing inputting outputting

Software Engineer

Wunderman Thompson Data

September 2021

February 2021

- Refactored RESTful API Netflix's Hystrix to SWF (Spring Web Flow) directed onto its Spring Bean Rest-Controller to transition database logic of billions of transactions
- Resolved NCOA (National Change Of Address) discrepancies between millions of records of PostgreSQL customer-clients' contact points and Oracle DBMS personal identity
- Updated reporting logic to handle edge cases with timestamp and archive flags in runtime Groovy expression to return validated information on client outreach for Humana and other companies
- Streamlined data read-write with Java 8 array streams and lambda parameter-expressions to improve processing and memory performance as well as increase readability to prevent errors
- Dynamically ran Apache Groovy functions in real-time computing with JDKs (Oracle Java SDKs (Software Development Kits)) programs on the Java Platform SE (Standard Edition)
- Created Jenkins crumb issuer tokens for CSRF (Cross-Site Request Forgery) protection middleware and handling frontend to backend communications
- Resolved defects by researching Elasticsearch error, warning, and info Kibana logs with Regex (Regular Expression) generated from Java loggers running the dynamic Groovy expressions

Software Engineer

Black Hills IP

December 2020

September 2020

- Created a cost projection reports with Spring Framework Core Technologies and Spring REST (Representational State Transfer) to simplify client management and financing renewals
- Served backend Nginx web servers with Elastic Beanstalk into EC2 (Elastic Cloud Computing) instances to provide fault tolerance and reproducible virtual applications
- Handled HTTP 5XX response status codes by configuring OAuth2 (Open Authorization) configurations to access resources hosted by AWS EC2 (Elastic Cloud Computing)
- Deployed local Spring Boot web applications built with ORM (Object Relational Mapping) for patient data and IoC (Inversion of Control) with Java Beans for code reuse

Software Engineer

Carlson Wagonlit Travel

August 2020

November 2019

- Created full-stack website in Nuxt.js to sunset multiple desktop apps multi-millions of financial transactions to prevent maintenance and manual processing saving $360,000 annually
- Developed SPA (Single-Page Application) to transition from many MVC (Model-View-Controller) desktop applications to single MVVM (Model - View - ViewModel) in Vue (Vue.js)
- Reconstructed Microsoft's LDAP (Lightweight Directory Access Protocol) AD (Active Directory) into a cross-platform SSO middleware in Nuxt.js Axios simplifying user state management
- RESTed (Representational State Transfer) a/synchronous data with Express (Express.js) to Microsoft SQL Servers processing millions of T-SQL statements on Microsoft SQL servers
- Created Express.js REST API (Representational State Transfer Application Programming Interface) to be flexible with internal JSON (JavaScript Object Notation) requests
- Created a SSO (Single Sign-On) website in Vue (Vue.js) MVVM (Model-View-ViewModel) for fast components, dynamically responsive, and resource minimization UI (User Interface)
- Developed interactive and animated SPA (Single-Page Applications) including hamburger menus, navigation panels, and kabob options with Vue-Loader CSS (Cascading Style Sheets) modules
- Converted backend web server from Python 3 to Node.js for a unifying single programming language of server-side and client-side writing with “JavaScript everywhere” paradigm
- Proved POC (Proof of Concept) of transitioning from VB.NET desktop backend to Python 3 web applications in PyCharm by JetBrains to bring a simplified and singular deployment services
- Built backend web application and its API in Express (Express.js) to replace Python backend web application and VB.NET desktop application tools with a SSO (Single Sign-On) website
- SSR (Server-Sided rendered) routing navigation in Nuxt.js for easy navigation accessibility, growable scalable design, and session state management

Software Engineer

MTS Systems Corporation

September 2019

February 2019

- Developed website backend in C# (C Sharp) .NET Core Framework, and NuGet PMS (Package Management System) for reporting and diagnosing billions of data points into summaries
- UDP (User Datagram Protocol) connected our MTS monitor reporting to AWS cloud storage with and AWS CLI (Amazon Web Services Command Line Interface)
- Migrated our .NET Core Framework codebase SCM (Source Control Management) from Apache TortoiseSVN (Subversion) to Microsoft GitHub for ease of access and plugin compatibility
- Multithreaded subprocesses to process Robot Framework evaluations and trigger AWS Lambda emailing into SMTP (Simple Mail Transfer Protocol) updates and reports
- Pip installed PyPI (Python Package Index) for our Python PMS (Package Management System) packages including: Boto3, AWS CLI, RE, Win32Gui, and PyWinAuto
- Converted code from C# WinAppDriver on Visual Studio 2008 to Python 3 Selenium on PyCharm IDE (Integrated Development Environment) to reduce licensing costs and setup concerns

Software Engineer

UHG (UnitedHealth Group)

January 2019

September 2018

- Programmed array streams and lambda parameter-expressions in Java 8 Platform SE (Standard Edition) and Groovy to keep consistency for the custom client solutions
- Managed response handling from enterprise messaging systems widgets to JSP (Jakarta Server Pages) for clients' healthcare clinics and their centralized response team
- Collaborated to review the needs of over 52 million US external-payers to analyze data and performance to demonstrate medical clinicians had the correct data and responsive websites
- Reinforced Kanban standards to address setup requirements for DevOps automation Validated SOAP and REST calls performed by backend services
- Engineered Java services for custom websites to prevent $1.7 billion losses in malpractice costs

Software Engineer

Omnitracs

August 2018

July 2016

- Developed REST API (Representational State Transfer Application Programming Interface) endpoints for data points in the trillions of driver and vehicle statuses and performance values
- Tracked long haul trucking and last mile delivery with commercial CAN Bus (Controller Area Network) data from embedded I/O (Input / Output) ports into the C# (C Sharp) web application
- Stored client information into the ODS (Operational Data Store) and ETL (Extract Transform Load) toward the DW (Data Warehouse) to meet IFTA (International Fuel Tax Association) regulations
- Processed CEP (Complex Event Processing) vehicles' GML (Geography Markup Language) and drivers' HOS (Hours Of Service) into On-Prem (On-Premise) PostgreSQL database servers
- Searched for specific patterns and character sequences in XPath (XML Path Language) with Regex (Regular Expression) in DOM (Document Object Model) and partitioned databases
- Performed API requests in SOAP (Simple Object Access Protocol) with XML and REST (Representational State Transfer) with JSON in Postman IDE to verify responses
- Called AJAX (Asynchronous JavaScript And XML) with XHR (XMLHttpRequest) to the .NET ORM (Object Relational Mapping) that map to our RDBMS (Relational Database Management System)

Software Engineer

Best Buy

2015

Education

Bachelor of Science

Iowa State University

Computer Engineering